

Experimental Determination of Moment of Inertia

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1 Abstract

This report investigates the moment of inertia of rotating bodies using experimental measurements of angular velocity as a function of time. Linear regression was used to determine the angular acceleration from the measured data, and the results were compared for different experimental configurations. The aim was to examine how the distribution of mass affects the moment of inertia and to evaluate how well the experimental results agree with theoretical expectations.

2 Introduction

Moment of inertia is a physical property of anything which rotates, characterized by the resistance to changes in angular velocity. Since this affects how much force needs to be applied to accelerate or decelerate objects, knowing how to calculate and measure it is crucial for many applications. In this study several means of measuring the moment of inertia practically were explored and compared with theoretical values. The aim was to determine if this was a valid means of measuring the moment of inertia and how accurate of a method it was. A second goal of the experiment was to determine the effect of impulse on a rotating system to show with a practical example that angular momentum is conserved in systems like this while angular kinetic energy is not.

3 Theory/Background

To calculate with rotational movement and can not idealize the mass to a particle as often done in translational calculations, you have to use expresions giving a link between the mass of an object and how it is distributed [2, p. 339]

$I = \text{moment of inertia}$, different geometric forms will give diferent calculations about its centroid being the center of the distributed mass.

Examples of equations of moment of inertia

Equation for a circular plate

$$I_x = I_y = m \frac{r^2}{2} \quad (1)$$

[2, p. 342]

Equation for a circular ring

$$I_x = I_y = \frac{1}{2} m (r_1^2 + r_2^2) \quad (2)$$

[2, p. 342]

Using this link Newtons second law $\sum F = ma$ analogusly can be aplied for rotational movement [2, p. 335].

$$\sum \tau = I\alpha \quad (3)$$

τ equals torsion I equals moment of inertia and α the angular acceleration

To calculate Torque (τ) it is the distance to where the aplied Force perpendicular to the rotational center.

$$\tau = Tr \quad (4)$$

[1, p. 171]

T is the aplied force in Newton and r is the distance to the point of perpendicular aplied force.

Given an expresion for $\tau = Tr$ the two formulas can be joined to

$$Tr = I\alpha \quad (5)$$

In the case that the force is generated with a free falling weight the following substitutions can be made $\sum F = mg - T = ma$. a = the linear acceleration of the falling mass giving the expression.

$$mgr = I\alpha \quad (6)$$

When the centroid of an object and the axes of rotation do not align The Paralell axel Theorem (Steiners's Theorem) can be used to calculate the moment of inertia.

$$I_{total} = I_{cm} + mr^2 \quad (r = \text{the distance between axis}) \quad (7)$$

[1, p. 181]

With the given equations a expression for the experimantal moment of inertia can be derived.

$$I = \frac{mgr}{\alpha} - mr^2 \quad (8)$$

For calculations of angular mommentum L and rotational energy k_{rot} fomulas analogous to straight movement are used.

$$E_{rot} = \frac{1}{2}\omega I \quad (9)$$

$$L = I\omega \quad (10)$$

[1, p. 181]

For error estimation the standard deviation can be calculated to give a sense of how close to the mean value the mesurements are distributed. x_i is the experimental value, $\bar{x}$ the mean value and n the amount of data collections

$$S = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}} \quad (11)$$

[1, p. 463]

4 Method

Two different experiments were conducted one to experimentally decide the moment of inertia and one to see what happens with angular momentum and rotational energy after an impact with an additional body following along with the rotation.

Both experiments were conducted using the PASCO Capstone system with the sensor setup shown in figure 1

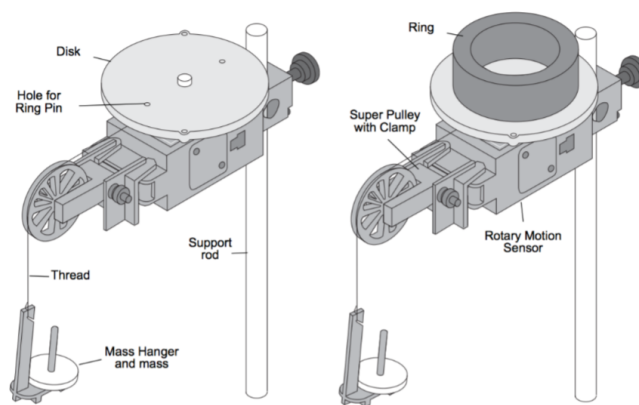


Figure 1: Graphic showing the sensor setup system.

the data was logged with angular velocity and time on the y and x axis in the Pasco system. to have controled force aplied in the system a smal weight atached to the disk in the sensor system was used. The weight was realesed and the PASCO system recoded the event, for the first experiment the flyweel alone was measured five times and with an additional ring weight added to the system as in figure 1

5 Results

Entity	Value	Unit
m_{disk}	119,63	g
m_{ring}	457,26	g
m_{weight}	9,66	g
r_{disk}	47,5	mm
$r_{ring\ outside}$	27	mm
$r_{ring\ inside}$	38,5	mm
$r_{attachment\ point}$	23,75	mm

Table 1: Measurements for ring, disk and attachment point

fisk

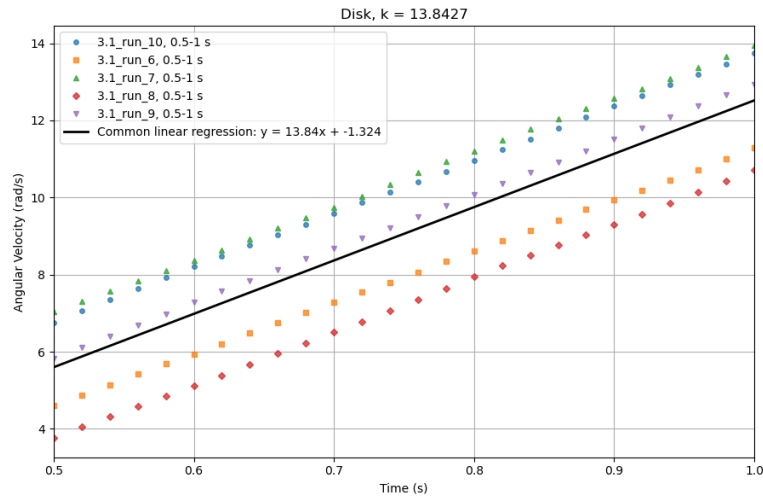


Figure 2: Graph from the experiment used to determine the moment of inertia of a disk. The k-value gives the angular acceleration.

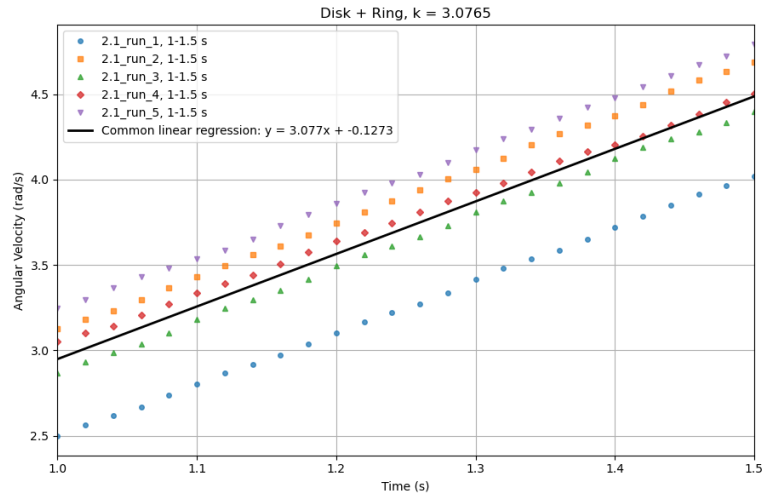


Figure 3: Graph from the experiment used to determine the moment of inertia of a disk with added weight. The k-value gives the angular acceleration.

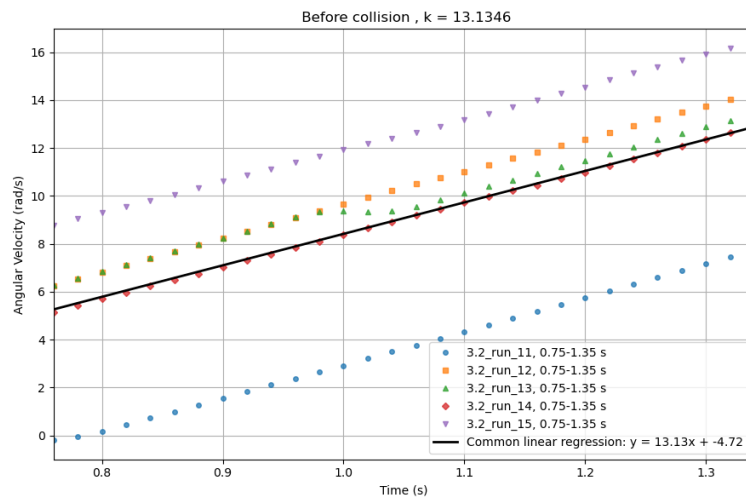


Figure 4: Graph showing the angular speed before impact of the added weight. The k-value gives the angular acceleration.

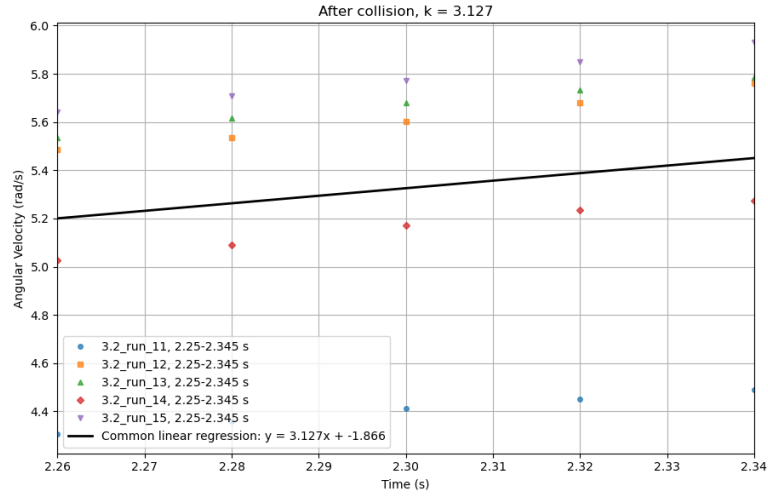


Figure 5: Graph showing the angular speed after impact of the added weight. The k-value gives the angular acceleration.

Experiment	Mean(ω)	Standard deviation
<i>Experimental I_{disk}</i>	14,04	0,089
<i>Experimental $I_{disk+ring}$</i>	3,022	0,0238
<i>Experimental $L/K_{pre-impact}$</i>	17,53	2,336
<i>Experimental $L/K_{post-impact}$</i>	4,095	0,498

Table 2: Standard deviation calculated according to equation: (11)

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6 Calculations

Using equation:(8) the experimental moment of inertia can be calculated only knowing the mass of the falling weight and the radius of the pulling disk.

$$I_{disk} = \frac{0,00966kg \cdot 9,81 \frac{m}{s^2} \cdot 0,02375m}{14,04 \frac{rad}{s^2}} - 0,00966 \cdot 0,2375m^2 = \quad (12)$$

$$I_{disk+ring} = \frac{0,00966kg \cdot 9,81 \frac{m}{s^2} \cdot 0,02375m}{3,022 \frac{rad}{s^2}} - 0,00966 \cdot 0,2375m^2 = \quad (13)$$

To calculate the theoretical values for the disk equation: (1) are used and for the disk and the ring it is combined with equation:(2)

$$I_{disk} = \frac{0,11963kg \cdot 0,0475^2m}{2} = \quad (14)$$

$$I_{disk+ring} = \frac{0,11963kg \cdot 0,0475^2m}{2} + \frac{0,455726kg \cdot (0,027^2m + 0,0385^2m)}{2} = \quad (15)$$

Before calculating the kinetic energy and the angular momentum, the angular moment has to be calculated, to calculate with the parallel theorem the mean displacement are used (6,4mm).

$$I_{before} = \frac{0,11963kg \cdot 0,0475^2m}{2} = \quad (16)$$

$$I_{after} = \frac{0,11963kg \cdot 0,0475^2m}{2} + \frac{0,455726kg \cdot (0,027^2m + 0,0385^2m)}{2} + \frac{0,19963kg \cdot 0,0064^2m}{2} = \quad (17)$$

To calculate and compare the Kinetic energy before and after the impact equation: (9) are used

$$E_{disk} = \frac{17,53 \frac{m}{s} \cdot I}{2} \quad (18)$$

$$E_{disk+ring} = \frac{4,095 \frac{m}{s} \cdot I}{2} \quad (19)$$

To calculate and compare the angular momentum before and after the impact equation: (10) are used

$$L_{disk} = I \cdot 17,53 \quad (20)$$

$$L_{disk+ring} = I \cdot 4,095 \quad (21)$$

7 Discussion

8 References

References

- [1] Carl Nordling and Jonny Österman. *Physics Handbook for Science and Engineering*. 9th ed. Studentlitteratur, 2020.
- [2] Hugh D. Young and Roger A. Freedman. *University Physics*. 11th ed. Pearson, 2003.

9 Appendix